

URBS

University Resource Borrowing System

A full-stack REST API system for managing and tracking university resource inventory and borrowing workflows

Node.js

Express.js

MongoDB

REST API

Cursor AI

SYSTEM OVERVIEW

URBS is a lightweight, full-stack web application that enables university students to request and borrow physical resources, while allowing administrators to manage inventory and approve or reject requests in real time.

PROBLEM

Universities lack a structured system for tracking borrowed items. Manual logs are error-prone, hard to audit, and create confusion around availability.

SOLUTION

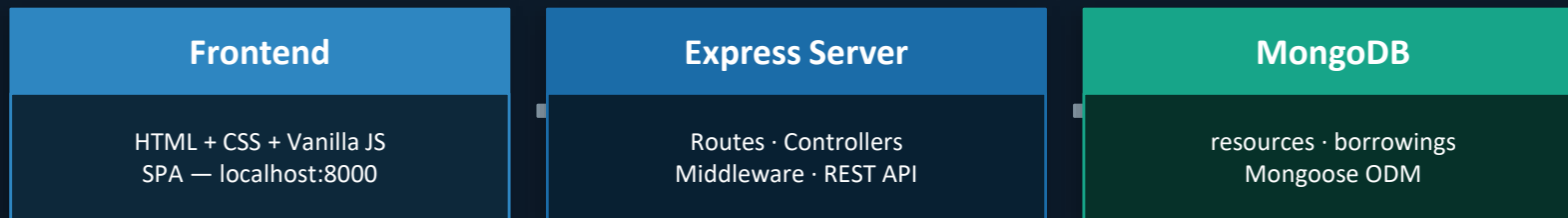
URBS provides a centralized REST API with a live web frontend — enabling digital inventory management and a traceable borrowing workflow.

USERS

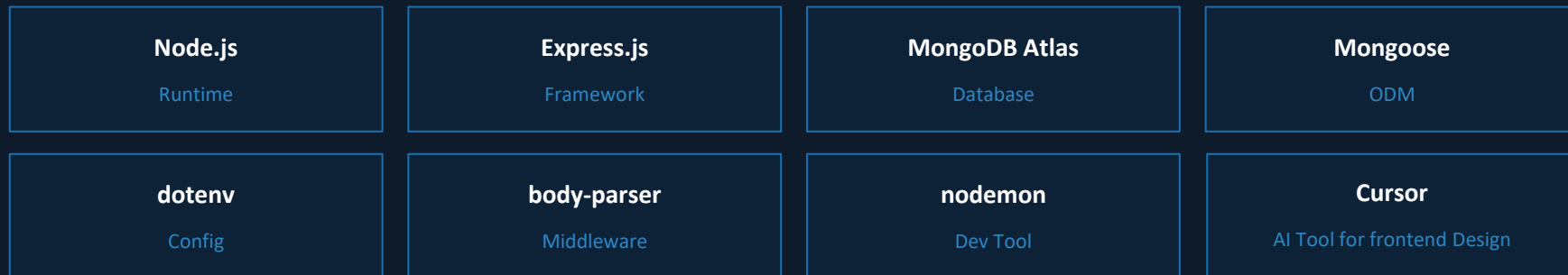
Students can browse resources and submit requests. Admins can manage inventory, approve or reject borrowings, and monitor overdue items.

TECH STACK & ARCHITECTURE

MVC Pattern | RESTful Architecture | Single-Page Frontend



DEPENDENCIES



DATA MODELS

Resource Model

name	String	Required
category	String	Enum (6 types)
description	String	Optional
totalQuantity	Number	Min: 1
availableQuantity	Number	Min: 0
condition	String	Enum (4 states)
location	String	Optional
addedDate	Date	Default: now

Borrowing Model

studentName	String	Required
studentId	String	Required
studentEmail	String	Required
resourceId	ObjectId	Ref: resources
quantityBorrowed	Number	Default: 1
expectedReturnDate	Date	Required
status	String	Enum (5 states)
adminNotes	String	Optional

API ENDPOINTS — Resources

Base: </api/resources>

POST

`/api/resources/add`

Add a new resource to inventory

GET

`/api/resources/getall`

Retrieve all resources

GET

`/api/resources/get/:id`

Fetch a single resource by ID

GET

`/api/resources/category/:category`

Filter resources by category

PUT

`/api/resources/:id`

Update resource details

DELETE

`/api/resources/:id`

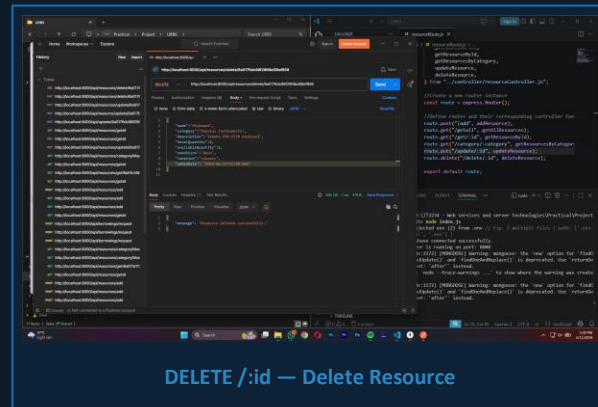
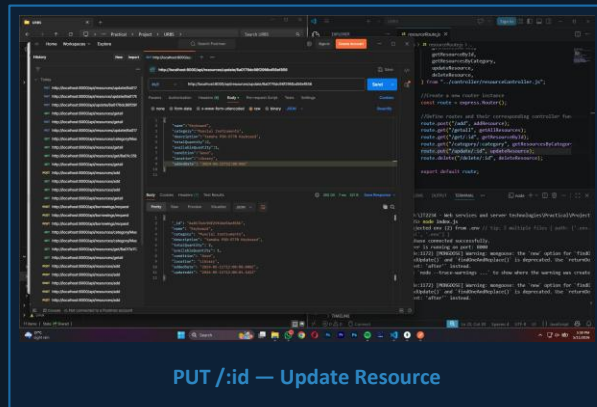
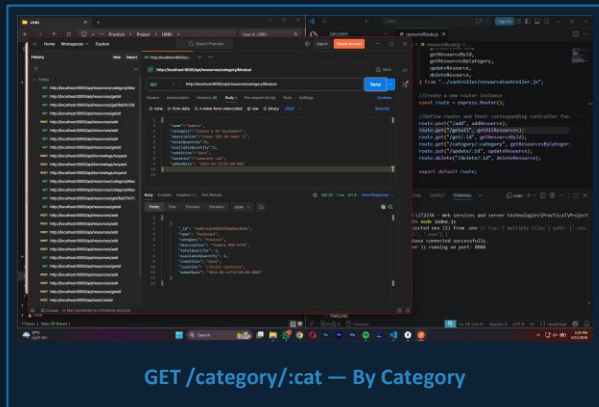
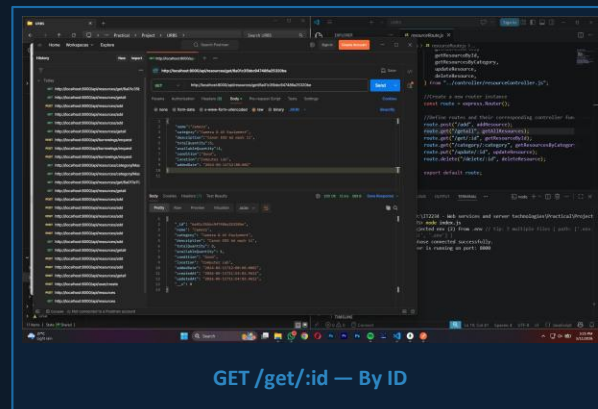
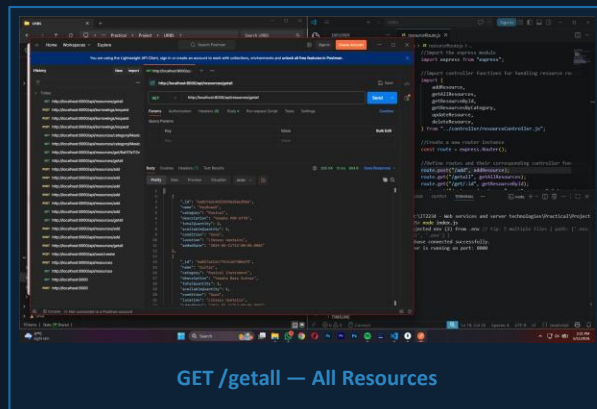
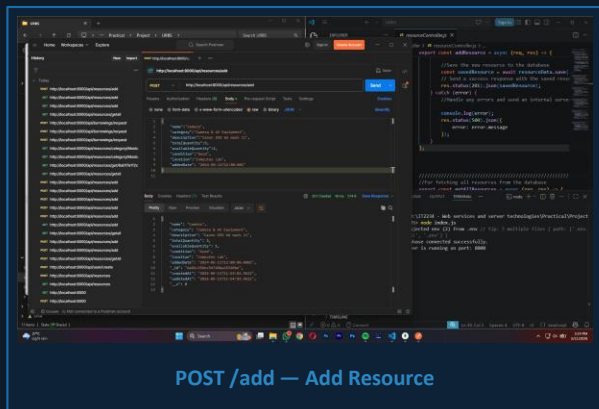
Delete a resource from inventory

API ENDPOINTS — Borrowings

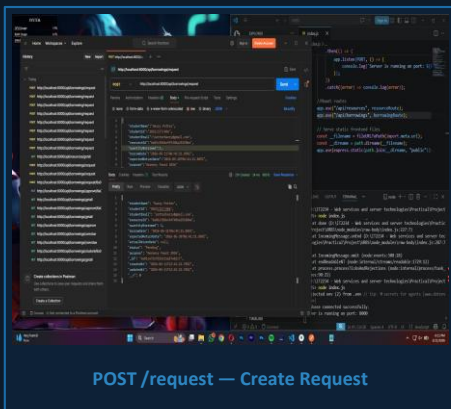
Base: </api/borrowings>

POST	<code>/api/borrowings/request</code>	Student submits a borrowing request
GET	<code>/api/borrowings/getall</code>	Admin retrieves all borrowing records
GET	<code>/api/borrowings/student/:studentId</code>	Fetch borrowings by student ID
GET	<code>/api/borrowings/overdue</code>	Get all overdue borrowings (auto-marks)
PUT	<code>/api/borrowings/approve/:id</code>	Admin approves a pending request
PUT	<code>/api/borrowings/reject/:id</code>	Admin rejects a pending request
PUT	<code>/api/borrowings/return/:id</code>	Confirm item return & restore quantity
DELETE	<code>/api/borrowings/:id</code>	Delete a borrowing record

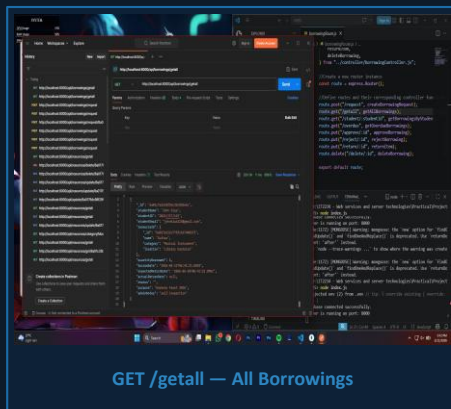
API TESTING — Resource Endpoints (Postman)



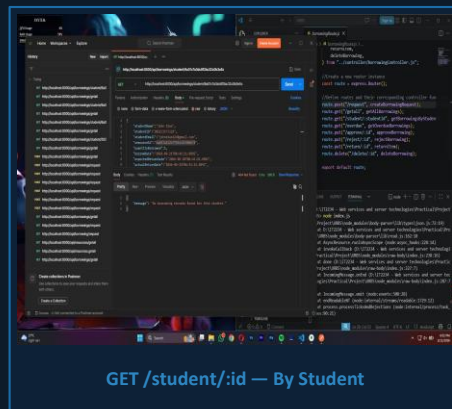
API TESTING — Borrowing Endpoints (Postman)



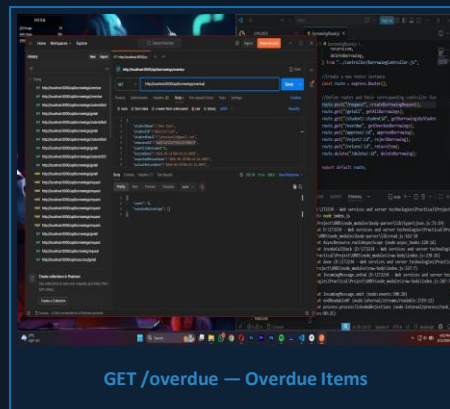
POST /request — Create Request



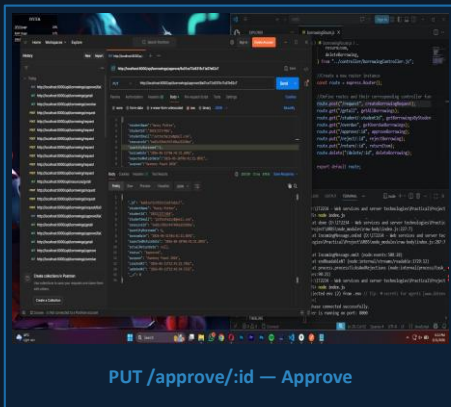
GET /getall — All Borrowings



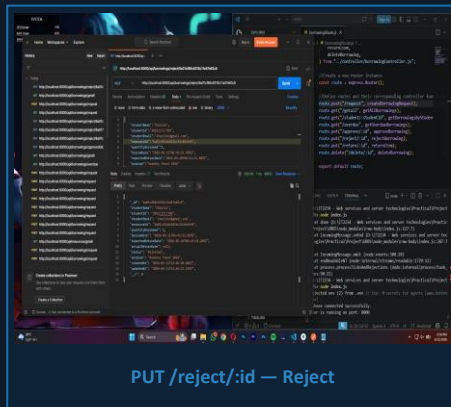
GET /student/:id — By Student



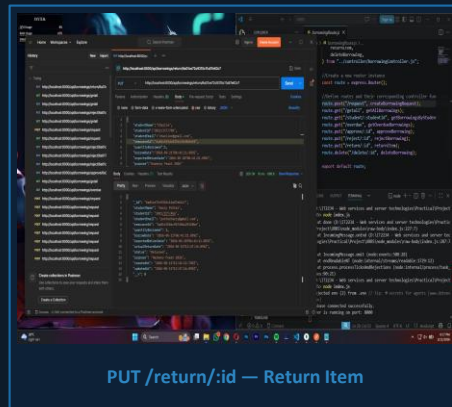
GET /overdue — Overdue Items



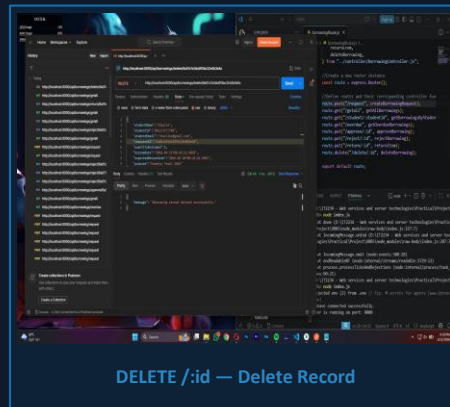
PUT /approve/:id — Approve



PUT /reject/:id — Reject

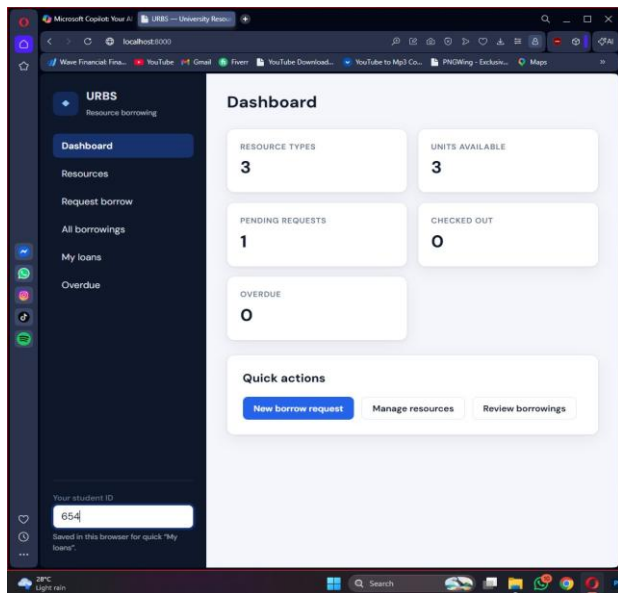


PUT /return/:id — Return Item

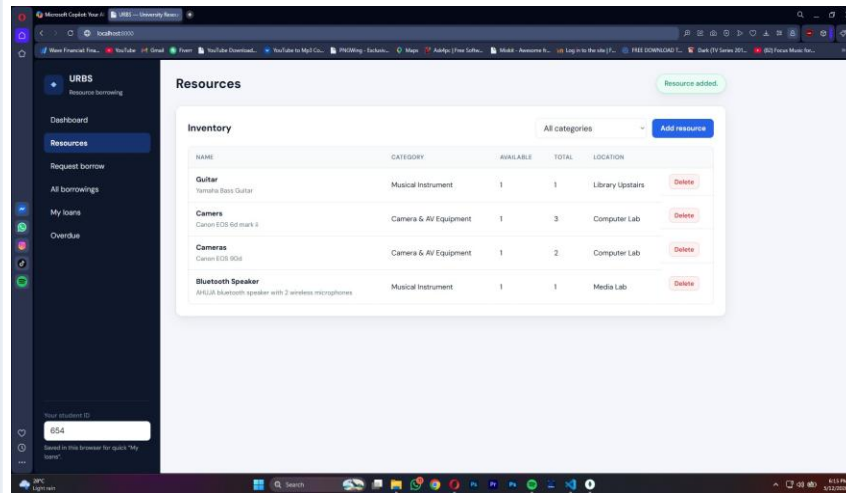


DELETE /:id — Delete Record

FRONTEND UI — Dashboard & Resource Management



Dashboard — Real-time Stats & Quick Actions



Resources — Inventory Table with Category Filter

FRONTEND UI — Borrowing Workflow

The screenshot shows the 'Request borrow' form in the URBS application. The form is titled 'Request a resource' and contains the following fields:

- Student name: Kevin
- Student ID: 982
- Email: kevin@gmail.com
- Resource: Bluetooth Speaker — 1 available (Musical Instrument)
- Quantity: 1
- Expected return: 05/14/2026 06:16 PM
- Purpose (Optional): FAS Sports Week

A 'Submit request' button is located at the bottom of the form. The left sidebar shows the navigation menu with 'Request borrow' selected.

Request Borrow — Student Form

The screenshot shows the 'All borrowings' table in the URBS application. The table has a 'Refresh' button in the top right corner. Below the table title, there is a note: 'Approve or reject pending requests. Return marks items back in stock.'

STUDENT	RESOURCE	QTY	DUE	STATUS	ACTIONS
Harry Potter 2022/CT1456 - potterharry@gmail.com	Cameras Camera & AV Equipment - Computer Lab	1	May 20, 2026, 12:18 PM	RETURNED	Remove
Charlie 2022/CT1789 - charlie@gmail.com	Cameras Camera & AV Equipment - Computer Lab	1	May 20, 2026, 12:18 PM	REJECTED	Remove
Rowley 654 - rowley@gmail.com	Cameras Camera & AV Equipment - Computer Lab	1	May 14, 2026, 6:15 PM	PENDING	Approve Reject

The left sidebar shows the navigation menu with 'All borrowings' selected.

All Borrowings — Admin Panel (Approve / Reject)

KEY FEATURES



Inventory Management

Add, update, delete, and filter resources by category. Auto-tracks available vs total quantity.



Borrowing Workflow

Full lifecycle: Request → Approve/Reject → Return. Duplicate request prevention built in.



Overdue Detection

Automatically marks approved borrowings as Overdue when the expected return date is exceeded.



Input Validation

Server-side validation for all fields — future dates, stock limits, duplicate checks, enum values.



Live Frontend SPA

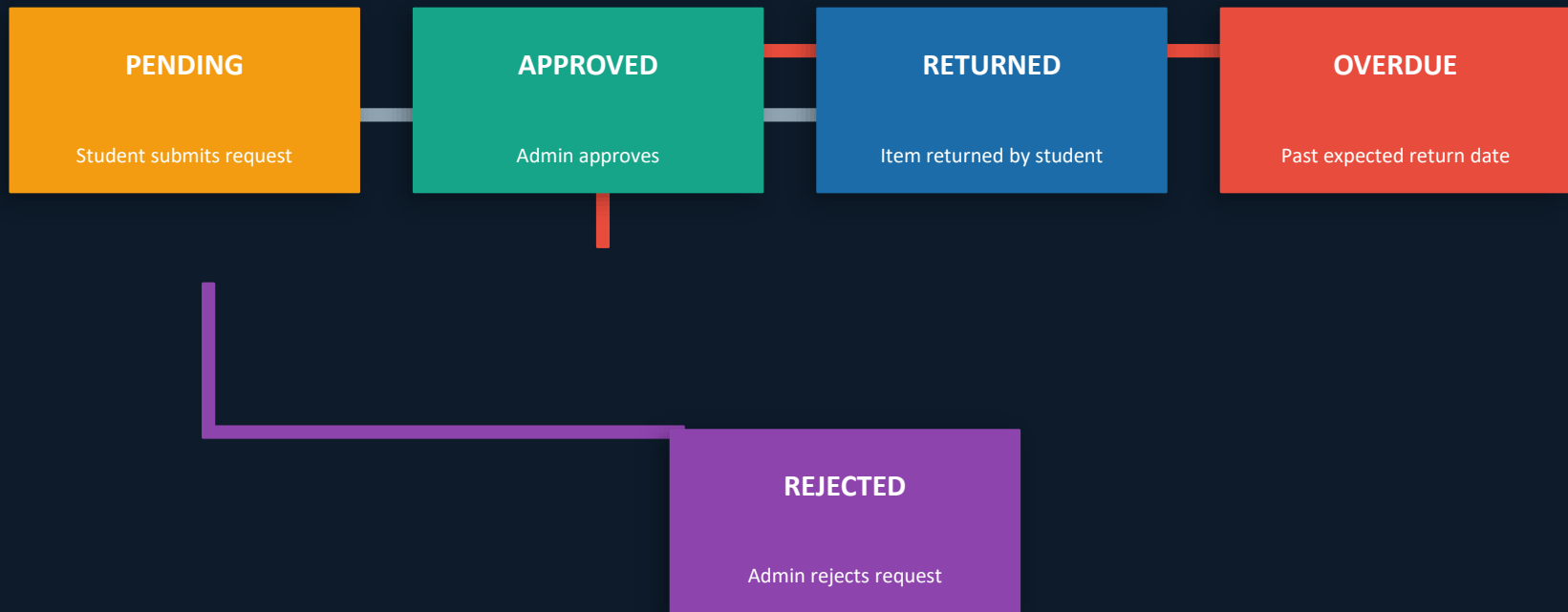
Single-page frontend with live dashboard stats, table views, filters, and modal forms.



RESTful API Design

14 clean endpoints. Proper HTTP methods, status codes, and JSON responses throughout.

BORROWING STATUS LIFECYCLE



Quantity is auto-deducted on Approve and restored on Return · Overdue is auto-detected on GET /overdue

URBS

University Resource Borrowing System

URBS provides a complete, production-ready backend with a clean REST API and a responsive web frontend — fully validated, tested in Postman, and ready for real university deployment.

T h a n k Y o u

2022/ICT/34
P. L. N. S. Cooray